Beautified Is Not Forged: A Two-Direction Measurement of Face-Forgery Detectors under Skin Smoothing

Anonymous submission

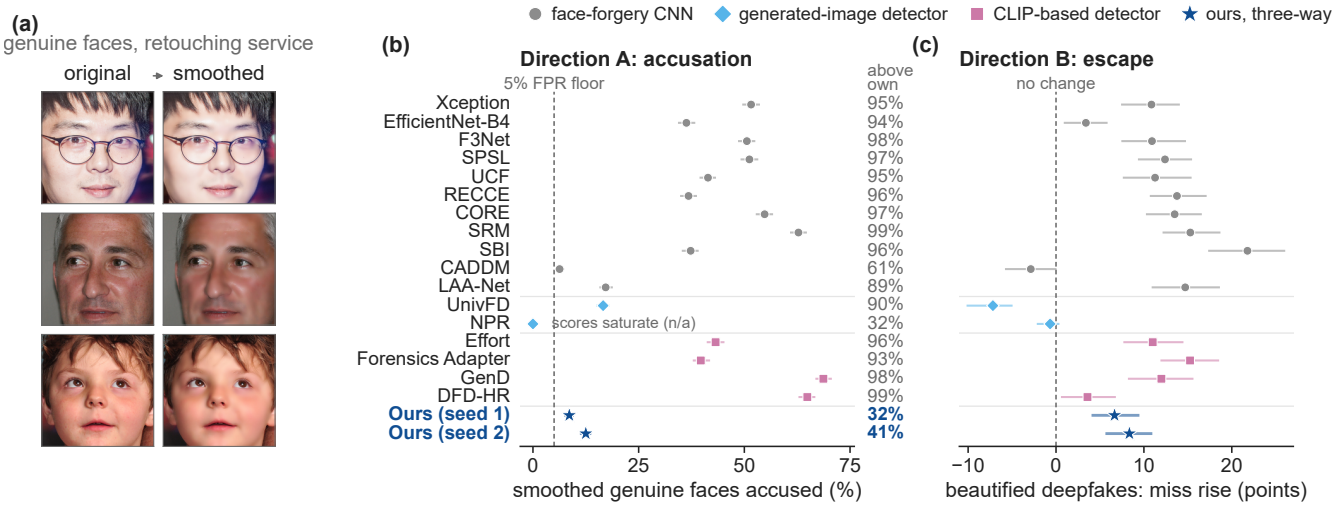


Fig. 1. The two directions of the measurement. (a) Three genuine FFHQ photographs and their renders by a commercial retouching service (skin smoothing, strongest level). (b) Direction A, one row per detector: share of the 2,210 renders accused at the threshold that accuses 5% of the untouched originals (dot, 95% interval; dashed line: that 5% floor); the grey number is the threshold-free reading, the share of pairs in which the render scores above its own original. NPR is n/a because its scores saturate on this source. (c) Direction B, same rows: rise of the miss rate on Celeb-DF-B deepfakes from the app presets alone, re-encoding subtracted (95% video-cluster intervals). Families from top to bottom: face-forgery CNNs, generated-image detectors, CLIP-based detectors, ours (two seeds).

***Abstract*—Published face-forgery detectors read skin smoothing as forgery evidence, and the error runs in both directions: a genuine face smoothed by a retouching service is accused, and a deepfake smoothed by a camera app is let through. We score seventeen published detectors, from Xception to the current CLIP-based detectors, on pixel-aligned retouching pairs and on app-beautified deepfake videos with one decision rule, and find that fifteen of them score the smoothed genuine face above its own original in 89–99% of pairs while the same detectors miss 3–22 points more of the beautified deepfakes than of the untreated ones once re-encoding is controlled for. Controlled training arms with architecture, data and recipe fixed show that the class which receives smoothed views during training sets the sign of the error, that exposure to smoothing is both necessary and sufficient for it, and that the cue is face-local, edge-preserving skin flattening rather than blur. A threshold cannot repair it, because under smoothing the scores of genuine and forged faces coincide; a consistency loss that removes the accusation turns out to be a smoothing-conditional recalibration, and every detector, including the fixed ones, loses separability when both classes are smoothed. A three-way detector trained with beautified views on both classes is the only model low on both axes: it accuses 9% of the smoothed genuine faces and keeps the highest AUC on beautified deepfakes, 0.87, while staying on par with the CLIP detectors on untreated forgeries. We release the sources, the scorer and the paired reader.**

I. Introduction

A person takes a selfie, lets the camera app smooth the skin, and uploads it. The face is the same person with the same expression at the same moment; only the skin texture has changed. If that photograph reaches a forgery detector during content moderation, a remote identity check or fact-checking, it should be called genuine. If a deepfake of the same person passes through the same app, it should still be called forged. Beautification edits the pixels a detector inspects but changes neither who is shown nor what the person does, and skin smoothing, whitening, eye enlargement and face slimming are default options in camera apps and retouching services[TODO: add a citable prevalence source for beautification before upload; none in the project files]. Fig. 1 shows what happens today: every published detector we scored ranks the smoothed selfie as more forged than the untouched one.

Three evaluations have looked at this. Libourel et al. [1] released Celeb-DF-B and report video-level AUC drops for three detectors under four app presets, without separating the two classes of error. Saeed et al. [2] apply six enhancement filters to deepfakes and measure how many escape six detectors;

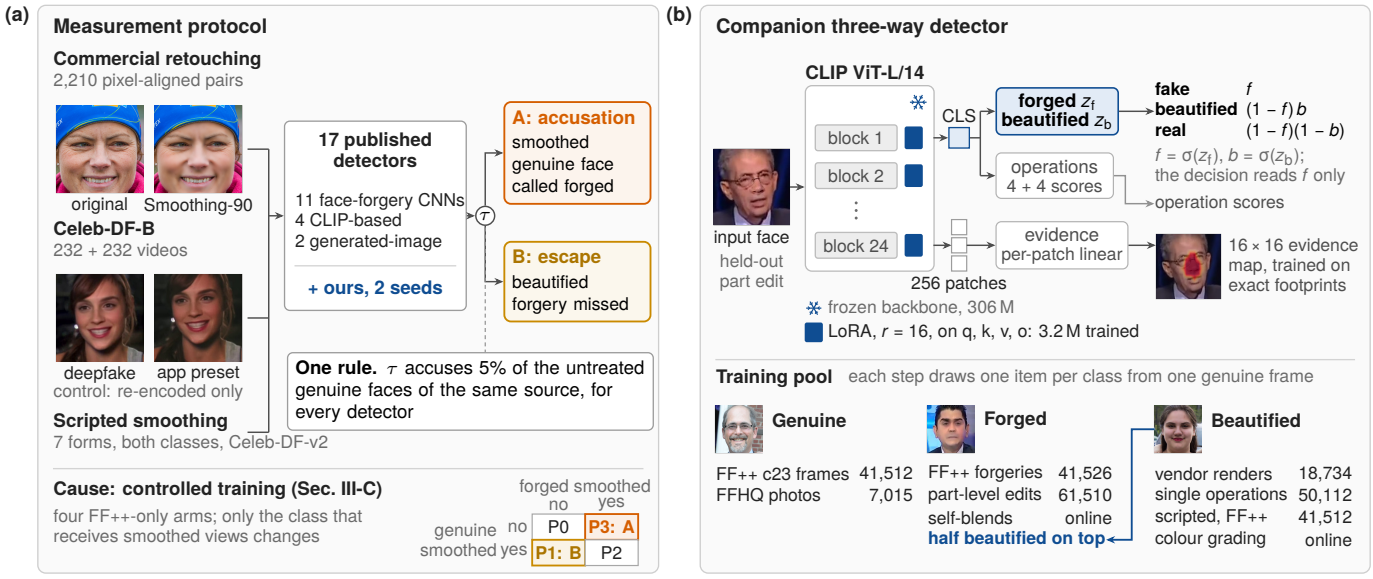


Fig. 2. Overview. (a) The measurement: beautified genuine and forged faces from three sources are scored by seventeen published detectors and ours, each thresholded at the score τ that accuses 5% of the untreated genuine faces of the same source; Direction A counts smoothed genuine faces called forged, Direction B beautified forgeries missed. The cause is tested with four training arms that differ only in which class receives smoothed views (Sec. III-C). (b) The companion detector of Sec. IV: a frozen CLIP ViT-L/14 with rank-16 LoRA, a forged and a beautified logit on the CLS token (the decision reads only f), auxiliary operation scores and a 16×16 evidence map on the patch tokens (shown on a held-out part-level forgery). Each training step builds one genuine, one forged and one beautified item from the same genuine frame; half of the forged items are beautified as well, so both classes see beautification.

genuine faces are never enhanced, so the accusation direction is not measured. Concas et al. [3] apply one smoothing filter to genuine and forged test images of two in-house CNNs and observe that the two detectors move in opposite directions, which they report but do not explain. Each study measures one direction, a handful of detectors, or no cause. In generated-image detection, Ricker et al. [4] showed that training with JPEG on one class only flips the sign of the error, with two of the four exposure cells. The implicit assumption shared by robustness benchmarks [5]–[7] is that a benign edit degrades a detector symmetrically, so that AUC under the edit is the number to report. We show that for skin smoothing the degradation has a sign, that the sign is set in training, and that AUC hides it.

We ask three questions. RQ1: across the detector families in use today, how often is a smoothed genuine face accused and how often does a smoothed forgery escape, under one rule? RQ2: where does the cue come from, which part of the edit carries it, and does it depend on what the detector was trained on? RQ3: can a decision rule or a training rule remove it without giving up detection of untreated forgeries?

A measurement that can answer these questions needs five properties. Both directions must be read on the same detector with the same rule. Where the untouched original of a beautified face exists, the pair should be read without a threshold, because a detector near saturation hides the shift behind its operating point. Re-encoding must be controlled, since every app re-compresses. The cause must be established by intervention on training, with everything else fixed. Uncertainty must be reported at the unit that was sampled, pairs or videos rather than frames.

We build such a measurement from six beautification sources, four of them never touched by a design decision, and score seventeen published detectors and a family of our own controlled arms with one rule. Our contributions are:

- A two-direction evaluation protocol and its results on seventeen detectors (Sec. II, Findings 1–2): on pixel-aligned commercial retouching pairs fifteen detectors score the smoothed genuine face above its own original in nearly every pair, and on app-beautified deepfake videos fourteen miss more forgeries once re-encoding is removed; smoothing the skin alone is enough.
- The cause by intervention (Findings 3–4): a complete 2×2 training-exposure design on face forgery, extending Ricker et al. [4] with the both-classes cell, shows that the class receiving smoothed views sets the sign of the error and that smoothing exposure is necessary and sufficient; a band-energy analysis shows that the cue is face-local and edge-preserving rather than loss of texture.
- Two negative results that bound what a fix can do (Findings 5–6): no threshold rule separates smoothed genuine from smoothed forged faces for any of the detectors, and a training rule that removes the accusation is a recalibration, since every model, fixed or not, loses separability when both classes are smoothed.
- A companion three-way detector (Sec. IV) that is low on both axes and on par with the CLIP detectors on untreated deepfakes, with its costs stated.

Identity as a reference channel is examined as a seventh, discussion-level finding because it is prior art.

II. Measurement Protocol

Classes. A face is *forged* if it shows a person, an expression, a motion or speech that was not captured: another identity in whole or in part, a reenactment, or a facial feature replaced by new content. It is *beautified* if its colour, tone, skin texture or the shape of the person's own features were edited after capture while identity, expression and pose stay those of the capture. The label follows what changed, not the tool, and a beautified forgery is forged. Cases near the boundary are listed in the supplement (Table S1); the boundary is stated, not measured.

One rule, both directions. Every detector is scored by its own $P(\text{fake})$ with the threshold that accuses 5% of each source's untreated genuine faces. Under beautification we report the false-alarm rate FA on beautified genuine faces (Direction A, the accusation) and the miss rate MISS on beautified forgeries (Direction B, the escape); a forgery that a three-way model sends to its beautified class counts as a miss. Where the untouched original of each beautified face exists we also report the share of pairs in which the beautified face scores above its own original, which needs no threshold and is the quantity of Fig. 1. Intervals are 95% percentile bootstraps over pairs or videos with the threshold re-estimated in every draw (1,000 draws), and differences between models use identical draws; a second bootstrap with identities as the unit changes none of the headline intervals (Table S14).

Sources. *Commercial retouching*: 2,210 FFHQ photographs [8] and their renders by one retouching service from RetouchingFFHQ [9], pixel-aligned, four operations at three levels; the strongest skin smoothing (Smoothing-90) is the headline condition, and the other operations serve as controls that pass through the same service and encoder. No model was trained on this service, but it was read during the study, so it is an analysis set rather than a blind test. *App-beautified deepfake videos*: Celeb-DF-B [1], 232 genuine and 232 forged Celeb-DF-v2 videos per condition (as recorded, re-encoded, beautified by one of four Instagram presets), 32 frames each; the rise in misses under a preset is read minus the rise under re-encoding, so what remains is the beautification alone. *Scripted smoothing*: 4,016 Celeb-DF-v2 test frames (1,348 genuine, 513 videos) under seven forms applied to both classes: a whole-crop bilateral filter at three strengths, a 5×5 median, a 50% Gaussian blend, and a bilateral filter confined to the landmark face hull at two strengths (*face-local*, background untouched). *Calibration renders*: FF++ validation genuine frames under the same forms, used only to fit the threshold rule of Finding 5. *Untreated detection*: Celeb-DF-v2 [10], DFD [11] and DF40 [12]. Counts, crops and preprocessing are in the supplement.

What is not a clean source. Instagram AR filters rendered through OpenFilter [13] (FairBeauty, B-LFW) were scored but are not used as evidence. The AR engine re-renders the face through its own capture path, and there is no no-filter pass through that path, so the filter cannot be separated from the pipeline: a model trained on such pairs calls 94.8% of filtered images beautified with the face greyed out, and the FairFace originals themselves are called genuine by only 18.2% of our best model. Every detector, ours included, accuses 10–97% of these renders (Table S9); we report that as a limitation and make no claim on AR filters.

TABLE I

Direction A on 2,210 pixel-aligned retouching pairs. Above own: share of pairs in which the Smoothing-90 render scores above its own original (50 means no shift). FA: share of renders above the threshold that accuses 5% of the originals. Right: the same FA for whitening, face slimming and eye enlarging at their strongest level. Brackets: 95% bootstrap over pairs. NPR scores everything at 1.0. Ours: the three-way detector of Sec. IV, two seeds.

	Smoothing-90		FA, other operations (level 90)		
Detector	above own [CI]	FA [CI]	white	slim	eye
Xception	95.2 [94.3, 96.1]	51.6 [49.5, 53.7]	5.7	3.7	4.6
EfficientNet-B4	93.7 [92.6, 94.7]	36.3 [34.3, 38.4]	5.1	4.0	4.7
F3Net	97.8 [97.1, 98.3]	50.6 [48.5, 52.6]	5.4	4.2	4.4
SPSL	97.3 [96.7, 98.0]	51.2 [49.1, 53.3]	5.2	4.9	4.9
UCF	95.4 [94.5, 96.2]	41.4 [39.4, 43.3]	6.6	4.0	4.2
RECCE	96.5 [95.7, 97.2]	36.8 [34.8, 38.8]	5.3	4.1	4.5
CORE	96.6 [95.8, 97.3]	54.8 [52.7, 56.8]	5.7	4.4	4.8
SRM	99.3 [98.9, 99.6]	62.8 [60.8, 64.8]	4.5	3.8	4.7
SBI	96.2 [95.4, 97.0]	37.3 [35.2, 39.2]	5.4	6.2	5.5
CADDM	60.9 [58.8, 62.9]	6.3 [5.3, 7.4]	4.7	6.0	5.9
LAA-Net	89.2 [87.8, 90.5]	17.2 [15.7, 18.9]	5.2	4.8	4.8
UnivFD	90.4 [89.1, 91.7]	16.6 [15.0, 18.1]	6.2	5.7	11.7
NPR	31.8 [29.7, 33.6]	0.0 [0.0, 0.0]	n/a	n/a	n/a
Effort	95.6 [94.8, 96.5]	43.2 [41.1, 45.3]	11.1	9.6	16.8
Forensics Adapter	93.3 [92.3, 94.4]	39.7 [37.8, 41.9]	5.4	6.2	7.0
GenD	98.5 [97.9, 99.0]	68.7 [66.8, 70.7]	8.1	6.4	7.5
DFD-HR	99.0 [98.6, 99.5]	64.9 [62.8, 66.8]	7.7	8.0	9.6
Ours, seed 1	**32.5** [30.5, 34.4]	**8.6** [7.4, 9.7]	4.8	3.7	3.6
Ours, seed 2	41.3 [39.3, 43.3]	12.5 [11.1, 13.8]	–	–	–

Detectors. Eleven face-forgery CNNs (Xception [14], EfficientNet-B4 [15], F3Net [16], SPSL [17], UCF [18], RECCE [19], CORE [20], SRM [21], SBI [22], CADDM [23], LAA-Net [24]), four CLIP ViT-L/14 detectors (Effort [25], Forensics Adapter [26], GenD [27], DFD-HR [28]) and two generated-image detectors (UnivFD [29], NPR [30]), all with released weights, each on its own crop; released checkpoints reproduce their published Celeb-DF-v2 numbers within the margins listed in Table S3. NPR saturates at 1.0 on the retouching source and is reported as such. Table S4 sets this protocol against the prior evaluations of Sec. I feature by feature.

III. Findings

A. Direction A: the accusation

Table I reads each Smoothing-90 render against its own original. Fifteen of the seventeen published detectors score the render above the original in 89–99% of pairs, and the median $P(\text{fake})$ moves from 0.21 to 0.99 for SBI and from 0.23 to 0.92 for GenD (Table S20): a heavily smoothed genuine face is scored like a forgery whatever the threshold. The two exceptions are NPR, whose scores saturate, and CADDM, which reads a boundary rather than texture and shifts in 60.9% of pairs. At the threshold that accuses 5% of the originals, the published detectors accuse 17–69% of the renders (CADDM 6.3%, NPR 0%). The operation is smoothing. Whitening, face slimming and eye enlarging at their strongest level stay within

TABLE II

DIRECTION B ON CELEB-DF-B. AUC ON UNTREATED FRAMES AND ON FRAMES BEAUTIFIED BY THE APP PRESETS (FRAME LEVEL), FA ON BEAUTIFIED GENUINE FRAMES, AND THE RISE IN THE MISS RATE UNDER BEAUTIFICATION MINUS THE RISE UNDER RE-ENCODING ALONE, AT THE 5%-FPR THRESHOLD OF UNTREATED GENUINE FRAMES. BRACKETS: 95% BOOTSTRAP OVER VIDEOS. BOLD: BEST PER COLUMN AMONG DETECTORS THAT DETECT THE UNTREATED FORGERIES (NPR IS AT CHANCE AND NOT BOLDED).

Detector	AUC untr.	AUC beautified [CI]	FA_b	ΔMISS beauty only [CI]
Xception	0.811	0.737 [0.702, 0.773]	5.7	+10.9 [7.4, 14.1]
EfficientNet-B4	0.811	0.762 [0.729, 0.798]	7.8	+3.4 [0.9, 5.9]
F3Net	0.798	0.731 [0.697, 0.767]	5.4	+10.9 [7.4, 14.8]
SPSL	0.781	0.727 [0.690, 0.762]	3.4	+12.4 [9.3, 15.5]
UCF	0.841	0.777 [0.746, 0.808]	5.7	+11.3 [7.6, 15.4]
RECCE	0.785	0.723 [0.687, 0.758]	4.1	+13.8 [10.7, 17.1]
CORE	0.821	0.752 [0.718, 0.786]	4.0	+13.5 [10.2, 16.6]
SRM	0.789	0.733 [0.698, 0.768]	2.8	+15.3 [12.1, 18.7]
SBI	0.875	0.752 [0.718, 0.785]	1.8	+21.8 [17.3, 26.1]
CADDM	0.886	0.762 [0.730, 0.790]	13.8	−2.9 [−5.8, 0.1]
LAA-Net	0.889	0.698 [0.660, 0.737]	6.0	+14.7 [10.9, 18.7]
UnivFD	0.653	0.624 [0.579, 0.663]	7.6	−7.2 [−10.2, −4.9]
NPR	0.503	0.498 [0.453, 0.543]	3.4	−0.7 [−2.2, 0.4]
Effort	0.909	0.807 [0.778, 0.836]	5.4	+11.0 [7.7, 14.5]
Forensics Adapter	0.909	0.828 [0.801, 0.856]	7.9	+15.2 [11.9, 18.6]
GenD	0.924	0.807 [0.776, 0.836]	9.9	+12.0 [8.2, 15.6]
DFD-HR	0.938	0.838 [0.807, 0.864]	14.6	+3.6 [0.6, 6.8]
OURS, seed 1	0.938	**0.866** [0.841, 0.890]	9.5	+6.7 [4.0, 9.5]
OURS, seed 2	**0.941**	0.883 [0.861, 0.902]	6.7	+8.3 [5.6, 11.0]

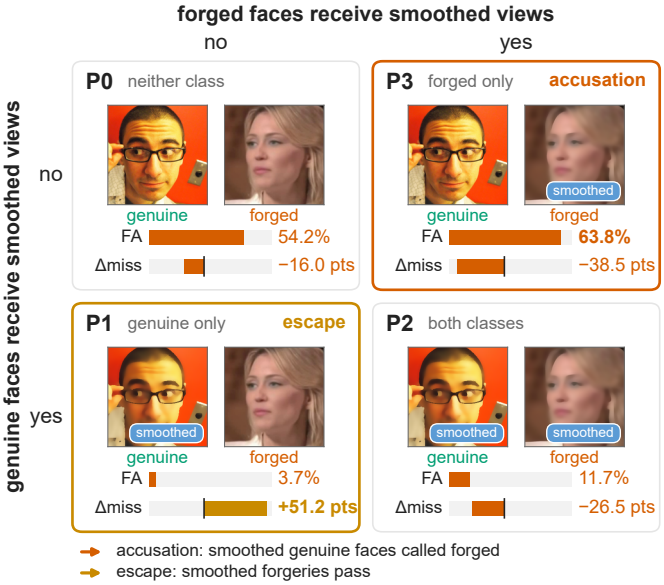


Fig. 3. The class that receives smoothed training views sets the direction of the error. Four FF++-only arms with the same architecture, data and recipe, first seed. FA: Smoothing-90 renders accused at the threshold that accuses 5% of their originals. Δmiss: change of the miss rate on Celeb-DF-v2 forgeries under strong whole-crop smoothing at the same rule; a negative change (red) means the arm accuses every smoothed face, a positive one (amber) that smoothed forgeries pass. P1 opens an evasion route; P3 accuses.

3.7–8.1% for thirteen of the sixteen detectors other than NPR although they pass through the same service, encoder and resolution.

Skin-only smoothing. A bilateral filter confined to the face hull, with the background untouched, is accused by all seventeen detectors (Table S12): the four CLIP detectors call 88–99% of genuine frames forged at the milder strength and 99.9–100% at the stronger one, the eleven CNNs 64–100% and 92–100%, UnivFD 14 and 55%. At matched visible strength skin-only smoothing is accused more often than whole-crop smoothing. For the four CLIP detectors the forgery logit rises in at least 97.9% of genuine frames at every strength, and the weakest face-local level already moves it by 1.0–1.9 times the detector's gap between its mean forged and mean genuine logit, the strongest by 1.9–3.0 gaps; a radial vignette alone moves three of the four the other way (Table S13).

Finding 1. *Every published face-forgery detector, whatever its family or year, scores a skin-smoothed genuine face as more forged than the same face untouched; smoothing the skin alone is enough, and the other beautification operations do not produce the shift.* The effect is not a threshold artefact: it appears in the pair ranking before any threshold is set. It holds for detectors trained on FF++ and for UnivFD, which was not, so it is not a property of one training set. Future evaluations of benign processing should read the pair before the AUC.

B. Direction B: the escape

Table II reads the same detectors on deepfake videos beautified by a real app. Here the accusation is small: beautified genuine frames are accused at 2–15%, close to the 5% floor for most detectors. The escape is not. Once the rise under re-encoding is subtracted, fourteen of the seventeen detectors miss 3–22 points more of the beautified forgeries, with the interval above zero; the largest rises belong to SBI, SRM and Forensics Adapter, and every detector that detects the untreated forgeries loses 0.04–0.17 of frame AUC beyond re-encoding (LAA-Net 0.870 to 0.698, SBI 0.861 to 0.752). Libourel et al. [1] reported the AUC drop for three of these detectors; the one-rule reading adds that the drop is almost entirely misses.

Finding 2. *The same detectors that accuse a smoothed genuine photograph let an app-beautified deepfake video through: beautification moves forgeries toward the genuine side by 3–22 points for fourteen detectors, and the loss is paid in misses rather than false alarms.* The two directions appear on different sources because the operating point differs: a retouching render lands above the threshold set on sharp originals, while an app preset applied to a forgery that was already scored near the threshold pushes it below. Both are the same shift read from two sides, which Finding 5 makes explicit.

C. The cause: training exposure

To find where the cue comes from we hold architecture, data and recipe fixed and vary only which class receives beautified views during training (Fig. 3, Table III). The arm without beautified views, P0, already accuses 54% of the Smoothing-90 renders, so an FF++-trained detector reads smoothing as forgery evidence out of the box. Giving the views to genuine faces only (P1) removes the accusation and opens an evasion route: on Celeb-DF-B its beautification-only miss rise exceeds P0's by 4.5–9.5 points in every one of three seeds (Holm-adjusted $p \leq 0.01$), and on frames it has never seen it misses

TABLE III

THE 2×2 EXPOSURE DESIGN. FF++-ONLY CLIP VIT-L/14 LORA ARMS, FORGERY HEAD ONLY, IDENTICAL EXCEPT FOR WHICH CLASS RECEIVES BEAUTIFIED TRAINING VIEWS (FACE-LOCAL SKIN SMOOTHING AMONG OTHER EDITS). SMOOTHING-90: RENDERS ACCUSED AT THE 5%-FPR THRESHOLD OF THEIR ORIGINALS [95% OVER ORIGINALS]. ΔMISS: CHANGE OF THE MISS RATE ON STRONGLY WHOLE-CROP-SMOOTHED CELEB-DF-V2 FORGERIES. CDF-B: BEAUTIFICATION-ONLY MISS RISE, THREE SEEDS. CDFV2: UNTREATED VIDEO AUC. P3 HAS ONE SEED.

Arm	smoothed views to	Smoothing-90 [CI]	ΔMISS	CDF-B ΔMISS, 3 seeds	CDFv2
P0	neither class	54.2 [50.5, 58.3]	−16.0	+17.0 / +14.2 / +10.8	0.938
P1	genuine only	3.7 [2.9, 4.4]	+51.2	+21.5 / +21.8 / +20.4	0.900
P2	both classes	11.7 [9.7, 14.3]	−26.5	+9.8 / +10.3 / +12.8	0.917
P3	forged only	63.8 [59.0, 67.8]	−38.5	–	0.943

96–100% of bilaterally smoothed forgeries when smoothing is the only view (a miss rise of 56–63 points on three seeds). Giving the views to forged faces only (P3) mirrors P1: it accuses 63.8% of the renders, 98–100% of smoothed genuine frames, and catches smoothed forgeries more often than untreated ones. Views on both classes (P2) sit in between, and on Celeb-DF-B the P2 minus P0 difference changes sign across seeds, so symmetric exposure is on average better than none but not reliably so (Table S15).

Two further arms show that smoothing is the operative view. Removing smoothing from the both-class views while keeping colour grades, filters, whitening and geometric edits brings the accusation back to 55–67% (exposure to smoothing is necessary), and smoothing as the only view on both classes lowers it to 2–5% (it is sufficient). The arms do not make the classes more separable; P3's untreated Celeb-DF-v2 video AUC is 0.943 against 0.938 for P0, and exposure moves errors between misses and false alarms.

Finding 3. *The class that receives smoothed views during training sets the sign of the error, and exposure to smoothing is both necessary and sufficient to set it: genuine-only views teach "smoothed means genuine" and open an evasion route, forged-only views teach the opposite, and no other beautified view substitutes.* This is shortcut learning [31] under class-dependent augmentation [32], which Ricker et al. [4] observed for JPEG with two cells; here all four cells are measured on face forgery, three of them with replication. The practical reading is a warning: a "filter-robust genuine class", the recipe a defender would reach for first, is the P1 cell.

D. The cue: face-local, edge-preserving flattening

If the detectors reacted to lost texture, the forms that remove more high-frequency energy would be accused more. They are not (supplement, Fig. S1): across seven scripted forms and seventeen detectors the false-alarm rate does not follow the energy removed above 0.25 cycles per pixel (pooled Spearman $+0.06$); the Gaussian blend removes the most energy and is accused least, the face-local filter removes less and is accused by nearly every detector. Three contrasts show what the detectors do follow. With the same kernel, confining the smoothing to the face raises the accusation by 28 points (16 of 17 detectors, $p = 5 \times 10^{-4}$); at equal energy loss an edge-preserving filter is accused 67 points more than a uniform blur (16 of 17); and only the band 0.02–0.12 cycles per pixel, the skin relief that a bilateral filter flattens and a Gaussian blend leaves, correlates with the accusation ($\rho = +0.45$).

TABLE IV

A SMOOTHING-CONDITIONAL THRESHOLD. FIXED: FAKE IFF $s > t_0$ WITH t_0 AT 5% FPR ON UNTREATED GENUINE FRAMES. CONDITIONAL: $t_0 + \Delta(b)$ WITH Δ LEARNT ON FF++ VALIDATION GENUINE RENDERS ONLY, b THE SMOOTHING-PRESENCE SCORE OF OUR THREE-WAY MODEL. FA ON FACE-LOCAL SMOOTHED GENUINE FRAMES (STRENGTH 0.12) AND MISS ON THE SAME SMOOTHING APPLIED TO FORGERIES, FIXED → CONDITIONAL. AUC: BOTH CLASSES SMOOTHED, SAME FORM. RECAL. MISS: MISS RATE WHEN THE THRESHOLD IS RE-SET AT 5% FPR ON THE SMOOTHED GENUINE FRAMES THEMSELVES (AN ORACLE). CELEB-DF-V2 FRAMES WITH LANDMARKS.

	FA (%)		MISS (%)		AUC	Recal.
Detector	fixed	cond.	fixed	cond.	both sm.	MISS
Xception	99.3	92.4	0.2	4.2	0.580	90.7
SBI	100.0	0.0	0.0	100.0	0.577	93.4
CADDM	99.9	0.0	0.0	100.0	0.572	91.7
Effort	99.9	0.0	0.0	100.0	0.658	84.0
Forensics Adapter	100.0	0.0	0.0	100.0	0.649	87.6
GenD	100.0	11.7	0.0	78.3	0.632	90.8
DFD-HR	100.0	0.6	0.0	99.3	0.601	93.5
OURS, forgery head only	84.5	0.1	2.5	100.0	0.755	73.7

Why would a detector trained on FF++ carry this cue? Inside the face, the Deepfakes and NeuralTextures forgeries of FF++ have less fine texture than their own source frames (high-band energy ratio -1.09 and -0.65, lower in 90–95% of videos), while FaceSwap is sharper ($+0.80$) and Face2Face unchanged. This is a correlation and a plausible origin rather than a demonstration: the low band that best predicts the accusation is not reduced in the training forgeries (at most -0.03, against -0.06 to -0.18 under our smoothing forms), and UnivFD, which never saw FF++, accuses skin-only smoothing too. An arm trained on Face2Face and FaceSwap alone, which would test the origin directly, is running and is not reported here.

Finding 4. *The shared cue is flattened skin with kept edges inside the face, which beautification produces and which forgery boundaries and generators leave behind; it is not blur, and texture loss as such does not predict the accusation.* This is why Gaussian blur or JPEG as a training view does not remove the accusation (Finding 6) and why any training-time fix has to use the smoothing signature itself.

E. A threshold cannot separate

A fixed threshold accuses smoothed genuine faces (Finding 1). The natural repair is to raise the threshold when smoothing is detected. We fit such a rule with the smoothing-presence output of our three-way model as the covariate and the shift learnt on FF++ validation genuine renders only, never on test data (Table IV; every detector in Table S28). It works on the genuine side: the accusation of strongly smoothed genuine faces falls to 0–1% for Effort, Forensics Adapter, DFD-HR, CADDM and our forgery head. It fails on the forged side at the same moment: 78–100% of the smoothed forgeries now pass. The reason is in the sixth column: with both classes smoothed, the AUC between smoothed genuine and smoothed

forged faces is 0.60–0.66 for the four CLIP detectors and 0.57–0.58 for SBI and CADDM, so the two score distributions overlap and any threshold that clears one clears the other. Even an oracle that re-sets the threshold on the smoothed genuine frames themselves leaves every published detector missing 67–100% of the smoothed forgeries at the two strong forms, and ours 52–77% (Table S26).

Finding 5. *No threshold rule separates smoothed genuine from smoothed forged faces for any of the detectors: under skin smoothing the two score distributions coincide, so the accusation and the escape are one distribution read from two sides, and a decision rule can only move the error from one side to the other.* Calibration on untreated data, the remedy proposed for smartphone photographs [33], does not apply here, because the problem is not where the threshold sits but that there is no threshold to find.

F. Training only trades

If the cue is set in training, can training remove it? We tested the rule that Finding 3 and Finding 4 suggest: one skin-smoothing operator drawn identically for every training item of every class, and a penalty λ on the change of the forgery logit between the clean item and its smoothed copy, with the clean logit held as a stop-gradient target. Bars were written before training (Table S24). Table V and Fig. S3 in the supplement give the family. Every penalty weight from 0.03 upward removes the face-local accusation (3–7% on smoothed genuine frames, against 52 and 85% for the reference), and the symmetric view alone ($\lambda = 0$) does not (54 and 92% at the two face-local strengths), so the penalty does the work. The commercial renders do not separate the arms: the reference already accuses only 9.1% of them and the penalty arms 2–13%. The cost has two faces that trade against each other along λ. At large weights the clean forgery score is pulled toward its smoothed copy and untreated detection falls (Celeb-DF-v2 video AUC 0.907 at $\lambda = 1$ against 0.956 without the rule); at small weights detection is kept (0.946 at $\lambda = 0.03$) but the smoothed forgery's score falls with the genuine one and its miss rises by 24 points. A one-sided penalty that forbids only a rise of the logit is the limit of that direction: untreated AUC 0.954 and 56 points more smoothed forgeries missed. A Gaussian blur of the same support in place of the smoothing operator fails the accusation bar at the strong face-local form (19.5%) and opens the bilateral evasion. DF40 reenactment is 0.06–0.09 below the reference at every weight, the one bar no member passes.

The penalty buys the operating point and leaves separability where the view put it. The control that settles this is the $\lambda = 0$ model read with the oracle smoothing-conditional threshold of Finding 5: with both classes smoothed it keeps an AUC of 0.832 (bilateral) and 0.764 (face-local) against 0.790 and 0.778 for SIC at $\lambda = 1$ and 0.802 and 0.778 at $\lambda = 0.1$, and its recalibrated miss rate (57 and 77%) is not higher than SIC's (64–73% on two seeds). SIC therefore equals a symmetric view plus a recalibration that one threshold can carry across untreated and smoothed inputs; it does not keep separation that the view alone would lose. Nor does a routed expert: a second model trained only on smoothed copies, selected by the smoothing score, clears the accusation (face-local 51 and 80% to 2 and 1%) and raises the miss on smoothed forgeries by 41–46 points, and its both-classes-smoothed AUC is below the untreated detector's on seven of the eight forms (one seed; the second is running; Table S27). The loss is general. With both classes smoothed, every one of the seventeen published detectors loses AUC, by 0.06–0.28 at the strong face-local form for the sixteen that are above chance, and the $\lambda = 0$ model, which saw smoothed views of both classes in training, still loses 0.061 and 0.129. Part of the forgery evidence is therefore erased by smoothing itself, beyond the shortcut; how much of the published detectors' loss is shortcut saturation and how much is erased evidence cannot be separated with these data, and we call the $\lambda = 0$ loss a measured loss, not a bound.

Finding 6. *Training can move the error but not remove it: a class-symmetric smoothing view with a consistency penalty takes the accusation of smoothed genuine faces from 85% to 4% at a cost in untreated detection, but it is a smoothing-conditional recalibration, since the view-only model read with a conditional threshold separates smoothed faces at least as well; with both classes smoothed every detector, published or ours, loses separability.* A fix for the accusation therefore has to be bought with misses on smoothed forgeries or with untreated detection, and a paper that reports only one of the three is reporting a point on this curve, not a solution.

G. Identity as a reference channel

Smoothing erases texture evidence but should not change who the person is. On the Celeb-DF-v2 probes of Finding 5, the ArcFace distance to the claimed person's other videos separates genuine from forged with an AUC of 0.996 untreated and 0.984, 0.963 and 0.982 under bilateral 15, bilateral 21 and face-local 0.12, against 0.924 falling to 0.823, 0.778 and 0.751 for our forgery head on the same frames (Table S29). Only its AUC is stable: at the untreated threshold the channel accuses 19, 40 and 21% of the smoothed genuine faces, because the genuine distance moves from 0.25 to 0.32–0.38 while the forged distance stays near 0.64, and its relative error $1 - \text{AUC}$ grows four to ten times, against two to four for the texture detectors. ICT and ICT-Ref [34], run with their released weights, lose 0.08–0.12 of AUC under the three stronger forms and fail the other way, letting smoothed forgeries through (Table S30), and fusing the identity distance with our forgery head is dominated by the distance alone at every condition, so it is not reported as a method. **Finding 7.** *A recognition embedding compared with a reference of the claimed person keeps most of its separation under skin smoothing, but only its AUC is stable; its operating point moves as much as a texture detector's.* Identity as detection evidence is prior art [34]–[36], the stable-AUC, moving-threshold pattern is known from face recognition under beauty filters [37], and the channel needs a reference and is blind to reenactment; we state it as a boundary for future detectors, not as a contribution.

TABLE V

SMOOTHING-INVARIANT CONSISTENCY (SIC) AGAINST THE SAME MODEL TRAINED WITHOUT IT (FORGERY HEAD ONLY, SAME DATA, VIEWS, SEEDS AND CHECKPOINT RULE). FA: SMOOTHED GENUINE CELEB-DF-V2 FRAMES CALLED FAKE AT THE 5%-FPR RULE (FACE-LOCAL 0.06 / 0.12, WHOLE-CROP BILATERAL 15); S-90: COMMERCIAL SMOOTHING-90 RENDERS ACCUSED; ΔMISS: MISS RISE ON FACE-LOCAL 0.12 SMOOTHED FORGERIES; UNTREATED VIDEO AUC ON CELEB-DF-V2 AND DFD; CELEB-DF-B AUC UNDER THE PRESETS; DF40 REENACTMENT VIDEO AUC (FF++ DOMAIN); BARS PASSED OF SIX PRE-REGISTERED BARS. THE VIEW-ONLY CONTROL ($\lambda = 0$) DRAWS THE SAME SMOOTHED COPY FOR BOTH CLASSES WITHOUT THE PENALTY.

Model	seed	smoothed genuine accused (%) fl 0.06	fl 0.12	bil 15	S-90 acc. %	Δmiss fl 0.12	untreated video AUC CDFv2	DFD	Celeb-DF-B AUC	DF40 FR ff	bars passed
forgery head only, no rule	1	51.6	84.5	35.5	9.1	−34.0	0.956	0.987	0.866	0.842	–
symmetric view only ($\lambda = 0$)	1	53.9	91.8	1.6	11.9	-40.1	0.943	0.975	0.891	0.835	3/6
SIC, $\lambda = 0.03$	1	5.0	3.3	9.6	11.2	24.4	0.946	0.979	0.886	0.750	4/6
SIC, $\lambda = 0.03$	2	3.8	3.5	8.8	15.0	20.4	0.933	0.975	0.876	0.777	2/6
SIC, $\lambda = 0.1$	1	6.5	4.2	9.9	10.5	15.8	0.930	0.980	0.871	0.774	3/6
SIC, $\lambda = 0.1$	2	4.6	4.6	8.5	8.1	17.6	0.929	0.974	0.868	0.771	2/6
SIC, $\lambda = 0.3$	1	6.0	4.2	8.1	12.8	9.5	0.898	0.971	0.840	0.692	2/6
SIC, $\lambda = 1$	1	5.8	4.4	5.8	2.1	9.7	0.907	0.986	0.835	0.768	2/6
SIC, $\lambda = 1$	2	6.2	5.5	6.3	12.3	6.1	0.844	0.968	0.804	0.746	2/6
one-sided penalty	1	0.1	0.1	0.0	9.2	56.5	0.954	0.980	0.880	0.794	4/6
one-sided penalty	2	0.1	0.0	0.0	10.4	47.3	0.929	0.975	0.860	0.775	2/6
Gaussian view (ablation)	1	5.8	19.5	1.1	3.5	-13.5	0.855	0.971	0.819	0.782	1/6
Gaussian view (ablation)	2	5.9	14.5	1.3	8.7	-9.3	0.848	0.960	0.821	0.784	2/6
SIC, $\lambda = 1$, FF++ only	1	7.4	6.8	5.3	7.0	4.8	0.856	0.979	0.829	0.674	2/6

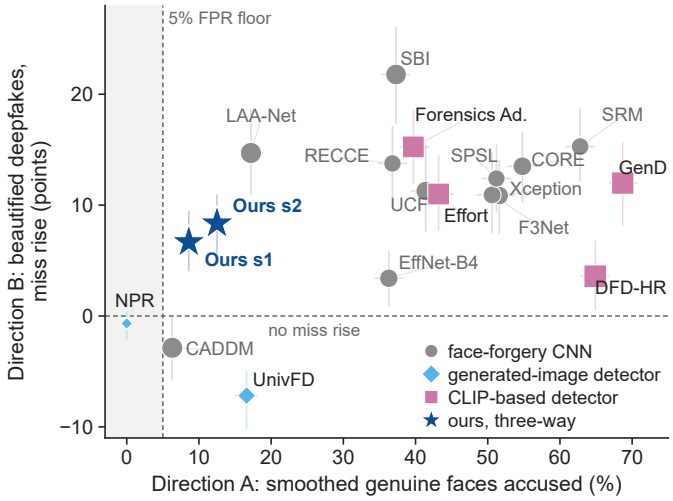


Fig. 4. Both directions at once. Each point is one detector: Direction A, Smoothing-90 renders accused at the threshold that accuses 5% of the originals (Table I), against Direction B, the beautification-only miss rise on Celeb-DF-B (Table II); bars are the 95% intervals, marker area grows with untreated Celeb-DF-B AUC, NPR is saturated at zero. The lower-left corner holds our two seeds and CADDM, whose interval contains zero but which accuses 13.8% of beautified genuine frames.

IV. A COMPANION THREE-WAY DETECTOR

The findings say what a detector that handles beautification must do: see smoothing on both classes in training, read the forgery decision from a head that is not asked to be smoothing-invariant, and report beautification as a separate attribute rather than as evidence either way. Our companion model does this (Fig. 2b).

System and training recipe. The backbone is a frozen CLIP ViT-L/14 [38] with rank-16 LoRA [39] on the query, key, value and output projections of all 24 blocks (3.2 M of 306 M parameters trained). Two linear heads on the CLS token give the forged logit z_f and the beautified logit z_b; with $f = \sigma(z_f)$ and $b = \sigma(z_b)$ the three classes are fake f, beautified $(1-f)b$ and real $(1-f)(1-b)$, and the decision reads $P(\text{fake}) = f$ only. An operation head (presence and amount of four operations) and a linear layer on the 256 patch tokens, which gives a 16×16 evidence map trained against the exact footprint of each edit, are trained alongside and are not read by the decision (supplement). Each training step builds three items from one genuine frame of FF++ c23 [40] or FFHQ: the frame itself, a beautified item (a landmark or skin operation at random strength, one of 14 colour filters [41], a random colour grading, or a render by two retouching services that are sister services of the test service, whole or decomposed into single-operation components) and a forged item (a self-blend [22], an observed FF++ forgery or one of 61,510 part-level forgeries), the forged item beautified with probability 0.5 so that both classes receive the views. FFHQ contributes 7,831 photographs (7,015 for training, 816 for checkpoint selection) and their renders. The test service itself is never seen. Training: 224 px, AdamW at 10^{-4}, 8 epochs; checkpoints are chosen on validation data only, by a balanced criterion that penalises originals sent to the beautified class (supplement).

Both axes. On the retouching pairs the model ranks the render above its original in 32.5 and 41.3% of pairs on its two seeds and accuses 8.6 and 12.5% at the 5%-FPR rule, with the paired difference against each of sixteen published detectors excluding zero (Table I); its native decision sends 98.8 and 99.2% of the renders to beautified and 0.8 and 0.6% to forged. On Celeb-DF-B it keeps the highest AUC under the presets, 0.866 and 0.883, with the paired interval against Effort, Forensics Adapter, SBI, GenD, DFD-HR and CADDM excluding zero on both seeds, and its beautification-only miss rise of 6.7 and 8.3 points is smaller than that of Forensics Adapter, SBI and GenD on both seeds and of Effort on one, but larger than that of DFD-HR and CADDM (Table II). It is

TABLE VI
UNTREATED DETECTION OF THE COMPANION MODEL AGAINST THE CLIP DETECTORS, EACH ON ITS OWN CROP. FRAME AND VIDEO AUC; BRACKETS: 95% PAIRED VIDEO-BOOTSTRAP INTERVALS OF OURS MINUS THE DETECTOR. EVERY INTERVAL CONTAINS ZERO.

	Celeb-DF-v2 frame	Celeb-DF-v2 video	DFD frame
Effort	0.879	0.927	0.933
Forensics Adapter	0.899	0.956	0.940
GenD	0.900	0.960	0.932
DFD-HR	0.886	0.951	0.942
OURS	0.891	0.947	0.942
minus Effort	[−0.013, +0.038]	[−0.006, +0.044]	[−0.006, +0.025]
minus Forensics Adapter	[−0.030, +0.013]	[−0.030, +0.010]	[−0.013, +0.017]
minus GenD	[−0.028, +0.009]	[−0.031, +0.003]	[−0.004, +0.024]
minus DFD-HR	[−0.014, +0.026]	[−0.022, +0.017]	[−0.012, +0.011]

therefore not the model with the smallest escape; it is the only model low on both axes, which Fig. 4 shows.

Untreated detection. On Celeb-DF-v2 and DFD the model is on par with all four CLIP detectors (Table VI); no paired interval excludes zero in either direction. On DF40 face swaps it is at Effort's level in the FF++ domain and 0.03 below it in the Celeb-DF domain (video AUC 0.952 and 0.905 against 0.944 and 0.935; no paired interval was computed).

Costs, stated. Three are real. On DF40 reenactment the model is behind: 0.718 and 0.502 in the two domains against 0.808 and 0.710 for Effort and 0.935 and 0.828 for GenD, and a forgery-head-only model on the same data is 0.08–0.15 above it, so the shared beautified head costs reenactment (Table S32). It sends 20.8 and 19.6% of the service's untouched originals to the beautified class, a cost the retouching source pays for the low accusation. Its four operation scores are reliable only for smoothing: on the service's single-operation renders at the strongest level the presence head recovers smoothing in 94% of cases but eye enlarging in 20%, face slimming in 32% and whitening in 38%, and on real-app filters that combine operations it recognises almost none of them (recall 0.003–0.24; Table S10). The operation names describe the training family rather than beautification in general, and we no longer claim them.

V. LIMITATIONS AND WHAT WE NO LONGER CLAIM

The measurement covers one commercial retouching service, four app presets on one video dataset, and seven scripted smoothing forms. The retouching source was read during the study and is an analysis set; a smoothing source that no design decision has touched is needed before the headline numbers can be called a blind test. Celeb-DF-B shares its genuine videos and identities with Celeb-DF-v2, on which every detector in the table was evaluated by its authors, so it is not identity disjoint. P3 and several SIC arms have one seed, and no non-CLIP pipeline was trained with the rule. Effort and DFD-HR were not reproduced at their published Celeb-DF-v2 numbers (Table S3), and we compare against their released weights.

Three claims of earlier drafts are withdrawn and one result is moved out. Instagram AR filters: the renders cannot be separated from the OpenFilter capture pipeline (Sec. II), our models accuse them as much as the published CLIP detectors (75–82% against 64–86%; Table S9), and training with multi-operation beautification on both classes does not lower the accusation (78 against 75%), so no statement about AR filters is made. Operation naming: the four operation scores identify smoothing on the training family and little else (Sec. IV). Language-model explanations did not differ from template sentences in any measured respect and were removed; localisation of part-level forgeries and its restoration test are in the supplement.

VI. RELATED WORK

Libourel et al. [1], Saeed et al. [2] and Concas et al. [3] are discussed in Sec. I; Libourel et al. close by recommending a dedicated filter-detection stage, which the companion model's beautified head is. Tariq et al. [6] benchmark fifteen detectors against moiré and name beautification as future work; von Stryk and Keuper [33] find that the false alarms of generated-image detectors on smartphone photographs follow threshold calibration, which Finding 5 shows is not available for skin smoothing. Retouching detection and restoration [9], [42]–[47] have no forgery class; OpenFilter [13] renders AR filters; Liu et al. [48] attribute editing tools in a closed set; perturbation benchmarks [5], [7] treat edits as corruptions of one class. Shortcut learning [31], group shifts [49] and class-dependent augmentation [32] describe the mechanism of Finding 3; Ricker et al. [4] found the sign flip for JPEG with two cells. CORE [20] ties representations across generic augmentations and still ranks a smoothed genuine face above its original in 97% of pairs; the alpha-blending hypothesis [50] holds that frame-based detectors read compositing artefacts, and our band analysis adds a second low-level cue. Identity-referenced detection [34]–[36] and face recognition under beauty filters [37] are the prior art for Finding 7; DeepfakeBench [51] and DF40 [12] provide the checkpoints and test sets.

VII. CONCLUSION

Seventeen published face-forgery detectors read skin smoothing as forgery evidence. Read with one rule in both directions, that single cue accuses the smoothed genuine photograph and lets the app-beautified deepfake through. The cue is set by which class saw smoothing in training, it is the face-local flattening that beautification produces, a threshold cannot remove it because the smoothed scores of the two classes coincide, and training can only move it along a measured trade-off between the accusation, the escape and untreated detection. A detector that sees smoothing on both classes and reports beautification as an attribute is low on both axes without losing untreated detection, at the costs we state. The sources, the rule and the paired reader are released so that the next detector can be read the same way.

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